IntelliCane: Finals Purchase & Power Engineering Guide

Neural-Nexus · Track A: Embedded IoT System · IoTrix 2.0 Finals (17 October 2026)

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1. Dual-MCU System Architecture (Safety Reflex + AI Vision)

To maintain the safety guarantees demonstrated in Phase 1 while adding real-time object identification for the finals, we employ a **two-microcontroller distributed architecture**:

- **Primary Safety Controller (ESP32-C3 SuperMini):** Handles safety-critical, 100% offline reflexes in a sub-50 ms loop: forward ultrasonic ranging, handle vibration PWM (200 Hz), silent fall/tilt detection via MPU6050, and Bluetooth Low Energy (BLE) telemetry to the smartphone. It never halts for network handshakes.
- **Secondary Vision Controller (ESP32-CAM + OV2640):** Acts as an on-demand AI camera coprocessor. Triggered when obstacles enter the 1 m warning boundary or upon user voice command. It captures JPEG snapshots and streams them via Wi-Fi to the companion Android phone for cloud/local object recognition and audio feedback.

2. Power Distribution Architecture & Brownout Prevention

The Engineering Challenge: The ESP32-CAM draws heavy transient current spikes of **450 mA – 600 mA** during Wi-Fi transmission bursts and camera sensor readout. If powered from a weak 3.3 V rail or an under-rated boost regulator, supply voltage sags below 4.5 V, causing the onboard AMS1117-3.3 regulator to drop out and trigger an immediate MCU brownout panic reset.

Power Supply Pipeline & Schematic Topology:

18650 Li-ion Cell (3.7 V, 2600 mAh) ⟶ **TP4056 (DW01A)** (Type-C Charger + Protection) ⟶ **SPST Switch** ⟶ **MT3608 Boost** (5.0 V 2 A) ⟶ **1000 μF Low-ESR Cap** ⟶ **5 V Common Rail**

- **Common 5 V Rail:** Both the ESP32-C3 (via `5V` pin) and ESP32-CAM (via `5V` pin) are powered from the boosted 5.0 V line. Their internal LDOs step this down cleanly to 3.3 V without mutual interference.
- **1000 μF Decoupling Reservoir:** A 1000 μF / 16 V Low-ESR electrolytic capacitor is soldered directly across the 5 V and GND pins of the ESP32-CAM. This acts as an immediate local charge reservoir, buffering 500 mA RF bursts and preventing voltage dips.
- **Power Budget & Runtime:** Continuous current is $\approx$ 400 mA @ 5 V ($\approx$ 2.0 W). A single genuine 2600 mAh cell provides $>$**4.0 hours** of continuous streaming and guidance. Putting the camera into Light Sleep between scans extends operation to $>$**8 hours**.

3. Materials Checklist: Vision & Power Subsystems

#	Component Description	Target Model / Specification	Qty	Est. LKR	Crucial Notes & Sourcing
Category A: Vision Subsystem					
1	ESP32-CAM Board	AI-Thinker ESP32-CAM + OV2640 2MP	1	2,400	Confirm 2MP camera ribbon cable is included in pack.
2	ESP32-CAM-MB Shield	CH340 Micro-USB Programmer Baseboard	1	950	**Crucial:** Snaps underneath; enables direct USB flashing.
Category B: Battery & Power Subsystem					
3	18650 Li-ion Battery	3.7 V 2500–3000 mAh (Genuine high-drain)	1	1,500	Reputable brand (e.g. LG, Samsung, or tested cell).
4	18650 Battery Holder	Single-cell holder with heavy solder leads	1	200	Secure physical cradle for the battery.
5	TP4056 Charger Module	Type-C with DW01A protection (`B+`, `B-`, `OUT+`, `OUT-`)	1	300	Must have 4 output pads for complete short-circuit safety.
6	DC-DC Boost Converter	MT3608 2 A Adjustable Step-Up Module	1	350	Calibrate output trimpot to exactly 5.0 V before connecting!
7	Decoupling Capacitor	1000 μF / 16 V Low-ESR Electrolytic	2	80	Mandatory across camera 5 V/GND to stop brownouts.
8	Noise Filter Capacitor	100 nF (0.1 μF code 104) Ceramic	2	30	High-frequency RF switching noise suppression.
9	Master Power Switch	SPST Mini Rocker or Slide Switch ($>$1 A)	1	80	Physical hardware power cut-off.

3 (Cont.). Materials Checklist: Prototyping, Wiring & Mechanical Mounting

#	Component Description	Target Model / Specification	Qty	Est. LKR	Crucial Notes & Sourcing
Category C: Circuit Prototyping & Robust Wiring (No Breadboard)					
10	Double-Sided Perfboard	FR4 Fiberglass Plated Through-Hole (5×7 cm)	2	300	Rigid mounting platform for modules and terminal blocks.
11	Female Pin Header Strips	2.54 mm pitch single-row (40-pin breakable)	2	150	Socket ESP32-C3 and ESP32-CAM to prevent heat damage.
12	Screw Terminal Blocks	2-pin & 3-pin (2.54 mm or 5.08 mm pitch)	4	250	Secure screw connections for battery, switch, and motor.
13	Flexible Silicone Wire	26 AWG or 28 AWG Stranded (Red, Black, Blue)	2 m	300	Won't fatigue or snap during cane movement.
14	Heat Shrink Assortment	2 mm to 6 mm diameter tubing	1 pack	200	Clean electrical insulation over solder joints.
Category D: Cane Chassis & Temporary Fasteners (3D Enclosure Held Off)					
15	Mobility White Cane	4-section folding white cane or aluminum stick	1	2,800	Standard assistive mobility chassis (surgical pharmacy).
16	Cable Zip Ties	Small & medium black nylon ties	1 pack	200	Sturdy strapping for perfboard and battery along the shaft.
17	Heavy-Duty Foam Tape	3M VHB or heavy-duty foam mounting tape	1 roll	300	Vibration-damped mechanical bond between cane and board.

4. Estimated Budget Summary & Sri Lankan Sourcing Guide

- **Vision Subsystem (ESP32-CAM + MB Daughterboard):** LKR 3,350
- **Power Subsystem (18650 Battery + Holder + TP4056 + Boost + Caps + Switch):** LKR 2,540
- **Prototyping & Interconnects (Perfboard, Headers, Terminals, Silicone Wire, Heat Shrink):** LKR 1,200
- **Cane Chassis & Fasteners (Folding Cane + Zip Ties + Foam Tape):** LKR 3,300
- **Total Estimated Budget:** ≈ **LKR 10,390** (≈ LKR 2,600 per team member across Dulnith, Thenul, Chamith, and Suneth).

Where to Purchase in Sri Lanka (Fast 24–48 Hour Availability):

- **TRONIC.LK (Nugegoda / Online):** In-stock ESP32-CAM + MB baseboards, TP4056 Type-C with protection, MT3608 modules, 18650 holders.
- **LankaTronix (Maharagama / Online):** Perfboards, stranded silicone wires, terminal blocks, electrolytic capacitors, and slide switches.
- **First Cross Street, Pettah (Walk-in):** Great for picking up 18650 cells, switches, heat shrink, zip ties, and fasteners immediately.
- **Surgical / Medical Supply Stores (Maradana / Kandy / Colombo):** In-stock 4-section folding mobility white canes.

5. Critical Assembly Rules & Verification Steps

1. **Calibrate MT3608 Before Plugging Anything In:** MT3608 trimpots often ship set to 15–20 V. Connect the 3.7 V battery and turn the brass screw counter-clockwise while probing with a digital multimeter until the output reads **exactly 5.0 V**. Only then solder it to the MCU rails.
2. **Power ESP32-CAM Through 5 V Pin (Never 3.3 V):** Always feed the regulated 5.0 V rail into the `5V` pin. The onboard AMS1117 regulator requires at least 4.75 V input to produce a clean 3.3 V output for the camera sensor.
3. **Install 1000 μF Capacitor at the Camera Header:** Solder the capacitor right between the `5V` and `GND` pins of the ESP32-CAM socket (observing polarity: negative stripe to GND).
4. **Keep Onboard Flash LED OFF in Firmware:** The bright white flash LED on the ESP32-CAM is tied to `GPIO 4`. Keep it `LOW` in setup (`pinMode(4, OUTPUT); digitalWrite(4, LOW);`) so it does not draw 150 mA of parasitic heat.
5. **Common Ground:** Ensure the ESP32-C3 ground, ESP32-CAM ground, TP4056 ground, and boost converter ground are soldered into a unified star ground plane.